KU LEUVEN

# Geavanceerde Computerarchitectuur
# Project Introduction

**Goals, examples, and grading**

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DRAMCO – WaveCore – ESAT
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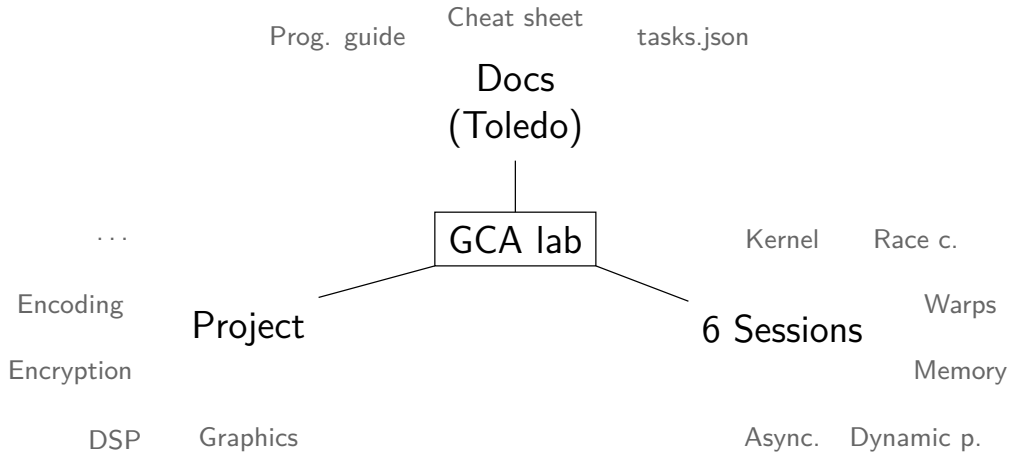
# 0 Outline

1. Recap from the first session
2. Project

# 1 Outline

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# 1 Organization of the lab and docs



# 1 Responsibilities and evaluation

## Lab seesions, 50 %

- Single-page report
- Single-page report
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- Single-page report

## Project, 50 %

- Project report
- Project Presentation
- Code release

# 2 Outline

1. Recap from the first session
2. Project

## 2 Goal and organization

- The goal of the project is to tackle a computational challenge using the GPU.
- This should be something that aligns with your interests – to support your thesis or another lab; to implement something for a hobby / private project; to boost your portfolio and get a better job.
- We will allocate 5 lab sessions to development and 1 session to the presentations.

# 2 Implementation

- Use the concepts we have gotten acquainted with so far and feel free to explore other paradigms. You can use another programming language and also build on another library – keep in mind that you should still understand and describe the intricacies of the implementation.
- At the lab, you can use the B230 PCs, an Nvidia Jetson Nano ("Raspberry with a GPU"), or the two Nvidia Teslas in the server. You are free to use other devices if you can/want.

# 2 Some topic examples

In case you don't know where to start, here are a few biased examples:

- An RF **ray tracer** to determine how Wi-Fi waves propagate in (and outside of) your kot.
- An **optimization algorithm** to determine where to put the Wi-Fi router for optimal coverage (considering either only yourself or all inhabitants).
- **Data compression** on an edge device that collects sensor samples and analyzes/compresses/transforms these in order to send the information over a low-throughput link.
- Real-time **statistical analysis** of a video stream.

# 2 Grading – guidelines based on thesis evaluation matrix

Implementation

- The implementation works, while clearly addressing the tackled problem
- Includes the concepts introduced in the lab or sufficiently motivates other approaches

Report (no min/max length – try to be concise)

- Clearly describes the problem and motivates the usage of the GPU
- Introduces the implementation and methods in detail (includes references to the repository or code listings)
- Highlights the shortcomings and possible improvements

Presentation (up to 10 min + 10 min for Q&A)

- Clearly describes the problem and motivates the usage of the GPU
- Provides a high-level overview of the implementation
- Highlights the shortcomings and possible improvements

# 2 Important dates

Please submit the reports on time, so we can close off the lab on the day of the presentations (focus on the thesis / exams afterwards).

The important dates for Monday's and Friday's group are:

- Project start – 25/03 and 29/03
- Hand in the reports – 09/05 and 20/05
- Presentations and finish – 13/05 and 24/05

Friday's group has a shifted deadline (longer gap) due to holidays.